1 Simplify fully $\dfrac{x^2 + 5x}{x^2 + 7x + 10}$

$$\frac{x\cancel{(x+5)}}{(x+2)\cancel{(x+5)}}$$

$\dfrac{x}{x+2}$

(Total for question 1 is 2 marks)

2 Simplify fully $\dfrac{x^2 - x - 12}{x^2 - 9x + 20}$

$$\frac{(x+3)\cancel{(x-4)}}{(x-5)\cancel{(x-4)}}$$

$\dfrac{x+3}{x-5}$

(Total for question 2 is 2 marks)

3 Simplify fully $\dfrac{3x^2 + 9x}{x^2 - 9}$

$$\frac{3x\cancel{(x+3)}}{(x-3)\cancel{(x+3)}}$$

$\dfrac{3x}{x-3}$

(Total for question 3 is 2 marks)

4 Simplify fully $\dfrac{x + 4}{x^2 - 16}$

$$\frac{1\cancel{(x+4)}}{\cancel{(x+4)}(x-4)}$$

$\dfrac{1}{x-4}$

(Total for question 4 is 2 marks)

5 Write $\dfrac{3x^2 + 11x - 4}{x^2 + 3x - 4}$ in the form $\dfrac{ax + b}{x + c}$ where a, b, and c are integers.

$$\frac{(3x-1)\cancel{(x+4)}}{\cancel{(x+4)}(x-1)}$$

$\dfrac{3x-1}{x-1}$

(Total for question 5 is 3 marks)

6 Write $\dfrac{x^2 + 7x - 18}{2x^2 - x - 6}$ in the form $\dfrac{x + a}{bx + c}$ where a, b, and c are integers.

$$\frac{(x+9)\cancel{(x-2)}}{(2x+3)\cancel{(x-2)}}$$

$\dfrac{x+9}{2x+3}$

(Total for question 6 is 3 marks)

7 Simplify fully $\dfrac{3x+6}{x-4} \div \dfrac{2x^2+9x+10}{x^2-4x}$

$$\frac{(3x+6)}{(x-4)} \times \frac{x^2-4x}{2x^2+9x+10}$$

$$\frac{(3x+6)(x^2-4x)}{(x-4)(2x^2+9x+10)}$$

$$\frac{3\cancel{(x+2)} \times x\cancel{(x-4)}}{\cancel{(x-4)}(2x+5)\cancel{(x+2)}}$$

$$\frac{3x}{2x+5}$$

$\dfrac{3x}{2x+5}$

(Total for question 7 is 3 marks)

8 Simplify fully $\dfrac{2x-2}{x+5} \div \dfrac{x^2-4x+3}{2x^2+13x+15}$

$$\frac{2(x-1)}{x+5} \times \frac{2x^2+13x+15}{x^2-4x+3}$$

$$\frac{2(x-1)}{(x+5)} \times \frac{(2x+3)(x+5)}{(x-3)(x-1)}$$

$$\frac{2\cancel{(x-1)}(2x+3)\cancel{(x+5)}}{\cancel{(x+5)}(x-3)\cancel{(x-1)}}$$

$\dfrac{2(2x+3)}{x-3}$

(Total for question 8 is 3 marks)

9 Solve $\frac{8}{x+3} + \frac{3}{x+8} = 1$

$$\frac{8(x+8)}{(x+3)(x+8)} + \frac{3(x+3)}{(x+3)(x+8)} = 1$$

$$\frac{8(x+8)+3(x+3)}{(x+3)(x+8)} = 1$$

$$8(x+8)+3(x+3) = (x+3)(x+8)$$

$$8x+64+3x+9 = x^2+8x+3x+24$$

$$11x+73 = x^2+11x+24$$

$$73 = x^2+24$$

$$0 = x^2-49$$

$$(x+7)(x-7) = 0$$

$$x = -7 \text{ or } 7$$

$x = -7$ or 7

(Total for question 9 is 4 marks)

10 Solve $\frac{8}{3x-2} + \frac{6}{x+1} = 2$

$$\frac{8(x+1)}{(3x-2)(x+1)} + \frac{6(3x-2)}{(3x-2)(x+1)} = 2$$

$$\frac{8(x+1)+6(3x-2)}{(3x-2)(x+1)} = 2$$

$$8(x+1)+6(3x-2) = 2(3x-2)(x+1)$$

$$8x+8+18x-12 = 2(3x^2+3x-2x-2)$$

$$26x-4 = 2(3x^2+x-2)$$

$$26x-4 = 6x^2+2x-4$$

$$0 = 6x^2-24x$$

$$0 = 6x(x-4)$$

$$x = 0 \text{ or } x = 4$$

$x = 0$ or 4

(Total for question 10 is 4 marks)

11 Solve $\frac{2}{5-x} + \frac{3}{x+7} = 1$

$$\frac{2(x+7)}{(5-x)(x+7)} + \frac{3(5-x)}{(5-x)(x+7)} = 1$$

$$\frac{2(x+7) + 3(5-x)}{(5-x)(x+7)} = 1$$

$$2(x+7) + 3(5-x) = (5-x)(x+7)$$

$$2x + 14 + 15 - 3x = 5x + 35 - x^2 - 7x$$

$$29 - x = 35 - 2x - x^2$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

$$x = -3 \quad x = 2$$

$x =$ -3 or 2

(Total for question 11 is 4 marks)

12 Solve $\frac{7}{x+1} - \frac{4}{3x-2} = 1$

$$\frac{7(3x-2)}{(x+1)(3x-2)} - \frac{4(x+1)}{(x+1)(3x-2)} = 1$$

$$\frac{7(3x-2) - 4(x+1)}{(x+1)(3x-2)} = 1$$

$$7(3x-2) - 4(x+1) = (x+1)(3x-2)$$

$$21x - 14 - 4x - 4 = 3x^2 - 2x + 3x - 2$$

$$17x - 18 = 3x^2 + x - 2$$

$$0 = 3x^2 - 16x + 16$$

$$(3x-4)(x-4) = 0$$

$$x = \frac{4}{3} \quad x = 4$$

$x =$ 4/3 or 4

(Total for question 12 is 4 marks)

13 Given that

$$2x+1 : x+2 = x+8 : 3x-4$$

Find the possible values of x.

$$\frac{2x+1}{x+2} = \frac{x+8}{3x-4}$$

$$(2x+1)(3x-4) = (x+8)(x+2)$$

$$6x^2 - 8x + 3x - 4 = x^2 + 2x + 8x + 16$$

$$6x^2 - 5x - 4 = x^2 + 10x + 16$$

$$5x^2 - 15x - 20 = 0$$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$x = 4$ $x = -1$

$x = 4$ or $x = -1$

(Total for question 13 is 4 marks)

14 Given that

$$x-1 : 2x-3 = x+2 : 3x-2$$

Find the possible values of x.

$$\frac{x-1}{2x-3} = \frac{x+2}{3x-2}$$

$$(x-1)(3x-2) = (x+2)(2x-3)$$

$$3x^2 - 2x - 3x + 2 = 2x^2 - 3x + 4x - 6$$

$$3x^2 - 5x + 2 = 2x^2 + x - 6$$

$$x^2 - 6x + 8 = 0$$

$$(x-2)(x-4) = 0$$

$x = 2$ $x = 4$

$x = 2$ or $x = 4$

(Total for question 14 is 4 marks)

15 Given that

$$x+9 : 5x-1 = x+7 : 2x-3$$

Find the possible values of x.

$$\frac{x+9}{5x-1} = \frac{x+7}{2x-3}$$

$$(x+9)(2x-3) = (x+7)(5x-1)$$

$$2x^2-3x+18x-27 = 5x^2-x+35x-7$$

$$2x^2+15x-27 = 5x^2+34x-7$$

$$0 = 3x^2+19x+20$$

$$0 = (3x+4)(x+5)$$

$$x = -4/3 \qquad x = -5$$

$x = -4/3$ or $x = -5$

(Total for question 15 is 4 marks)

16 Given that

$$5-3x : 9-x = 3x+7 : 4-x$$

Find the possible values of x.

$$\frac{5-3x}{9-x} = \frac{3x+7}{4-x}$$

$$(5-3x)(4-x) = (3x+7)(9-x)$$

$$20-5x-12x+3x^2 = 27x-3x^2+63-7x$$

$$3x^2-17x+20 = 20x-3x^2+63$$

$$6x^2-37x-43 = 0$$

$$(6x-43)(x+1) = 0$$

$$x = \frac{43}{6} \qquad x = -1$$

$x = \frac{43}{6}$ or $x = -1$

(Total for question 16 is 4 marks)